The Second Corpus Study on German Nominal Compounding Constraints

Background

German nominal compounding is a notoriously unpredictable and irregular process. About 40% of German N+N compounds insert a so-called linking element that is not confidently predictable for most of constituent classes. For example, nouns Birne ‘pear’ and Kirsche ‘cherry’ build compounds Birne-n-saft ‘pear juice’ but Kirsch-0-saft ‘cherry juice’ despite having nearly identical phonological, morphological, and semantic properties. The only linguistic generalizations available to gauge the correct linker are (mostly) irregular constraints that are eagerly violated by numerous compounds.

Our first corpus study on German nominal compounding collected 64 constraints from the linguistic literature and verified 39 of them against ~280k German N+N compounds. The corpus study demonstrated that most of the constraints were irregular, indeed. The most crucial funding was that not all of the constraints were confirmed on the massive data. Specifically, eight constraints were only partially supported by the data, and three were not supported at all.

The main limitation of the first corpus study was that none of the constraints involving semantic or lexical conditions could be implemented. The present study fills in most of the gaps remaining after the first corpus study:

1. It implements 17 constraints involving semantics. Fourteen of the 17 constraints are pluralic constraints actively discussed in recent linguistics literature.
2. It applies more detailed statistic analyses to analyze relations between different ontological properties of the constraints (application manner, types of linguistic conditions etc.) and its coverage and regularity measures.
3. It additionally analyses constraints through the Yang 2016's Tolerance Principle lense and makes a first attempt to discriminate between linguistic constraints and statistical patterns, and to model compounding with the Elsewhere Condition.

Additionally, it implements a couple of further constraints and verifies the constraints on further datasets.

Datasets

GeCoDB_v06

> 282767 compound lemmata. 3614 N1 lemmata. 9 linkers, 11 linker allomorphs. 4768 unique N1+linker combinations.

~280k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Database/compounds/decow16ax_comps_counts.txt'), with further processing steps in Shmalts BA Thesis, final preprocessing in Shmalts 2026.

The dataset is used to (re-)verify 40 non-semantic constraints.

GeCoDB_v07

> 275411 compound lemmata. 3593 N1 lemmata. 10 linkers, 12 linker allomorphs. 4741 unique N1+linker combinations.

~275k German N+N compounds from *GeCoDB_v06* whose constituents are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06*.

GeCoDB_v07_GN

> 48976 compound lemmata (48660 unique N1+N2 combinations). 3319 N1 lemmata. 10 linkers, 12 linker allomorphs. 3585 unique N1+linker combinations.

~49k German N+N compounds from *GeCoDB_v07* that are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06* and five further semantic non-pluralic constraints.

SP18_Pl_Constr

> 2681 compound lemmata. 44 N1 lemmata. 8 linkers, 9 linker allomorphs. 78 unique N1+(Sg/Pl meaning) combinations. 88 unique N1+linker combinations.

~2.5k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo, with annotated singular/plural meaning of N1. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Corpusstudy/Concordance.full.csv').

The dataset is used to verify 12 pluralic constraints.

Outline

1. Chapter 1. We first inspect the ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language: application manner, type(s) of accessed linguistic properties, application item. The results corroborate and slightly elaborate the finding from the first corpus study.
2. Chapter 2. We then inspect the impact of data and metric choice on surface constraint applicability. We find that the constraint applicability values are statistically very similar both across the *GeCoDB* dataset versions and the item/type variants of the metrics. Based on these results, we focus on the results of *GeCoDB_v07_GN* and *SP18_Pl_Constr* and only concentrate on the item variant of the applicability statistics.
3. Chapter 3. We analyze the values of application measures of the constraints on the flattened applicability statistics, which includes their distribution and correlation. We establish relation between the values of coverage and regularity of the constraints, and their ontological properties from step 1.
4. Chapter 4. We finally introduce and apply the new (irregularity) scale to our results.

Results

1. Constraint Ontological Properties

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. Al but two out of 65 constraints do not access the linguistic properties of N1.

2. Constraint Applicability Vs. Datasets & Metrics

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated across all pairwise comparisons of *GeCoDB* versions. That speaks in favor of the validity of the original results. Additionally, it attests the sufficient data quality and ensures the robustness of the constraint applicability statistics. We choose to proceed with *GeCoDB_v07_GN* as the dataset with the most verified constraints.
- The item and type variants of the applicability metrics are statistically very similar, which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

3. Constraint Applicability

- Most of the item coverage values are very low at <0.05. The item regularity values are more even. They are clustered naturally by the thresholds of 0.7 and 0.9, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$, $[0.7; 0.9)$, $[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints, while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho -0.184 and -0.342, respectively).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff -0.21 at intercept 0.85, p-value 0.001), while no such effect is observed for the other

domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

[4. Constraint Support, Productivity & Regularity]

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 are now partially supported. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*.
- Eleven constraints out of 49 turn out to be unproductive by the Tolerance Principle. Regular constraints by the RTP constitute slightly less than a half of the constraint set: 23 out of 49.
- We find that no constraints require a change of definition. Only regularity levels should be adjusted for them.

3. Constraint Applicability

Description

This chapter is the core part of the notebook. It analyses the applicability values of the constraints, which includes their distribution and correlation, and seeks to establish a relation between the values of coverage and regularity of the constraints, and their ontological properties considered in chapter 1.

1. First, we qualitatively analyze the distribution of the applicability values of the constraints. We observe a natural clustering of the item coverage and regularity values, and propose a new (ir)regularity scale based on it.
2. We compare the item coverage and regularity values of the P2L vs. L2P constraints. The L2P constraints appear to be statistically more regular. The difference in coverage is unexpectedly insignificant.
3. We test for a correlation between coverage and regularity. We find that there exists a small negative correlation between both item and type coverage and regularity.
4. We investigate the effect of the domain type of constraints on their item regularity. We find that semantics is the only domain that has a significant effect on item regularity. As expected, the effect is negative.

NB! In this chapted, we filter out all unconfirmed constraints (item regularity of $<0.5$$<0.5$) and constraints with $<3$$<3$ covered items.

Results

- Most of the item coverage values are very low at $<0.05$$<0.05$. The item regularity values are more even. They are clustered naturally by the thresholds of $0.7$$0.7$ and $0.9$$0.9$, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$$[0.5; 0.7)$, $[0.7; 0.9)$$[0.7; 0.9)$, $[0.9; 1.0]$$[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints, while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho $-0.184$$-0.184$ and $-0.342$$-0.342$, respectively).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

First, we only keep constraints that have at least $3$$3$ covered items and an item regularity of $\geq 0.5$$\geq 0.5$. That results in $49$$49$ constraints. Again, the number of the P2L constraints doubles the number of the L2P constraints.

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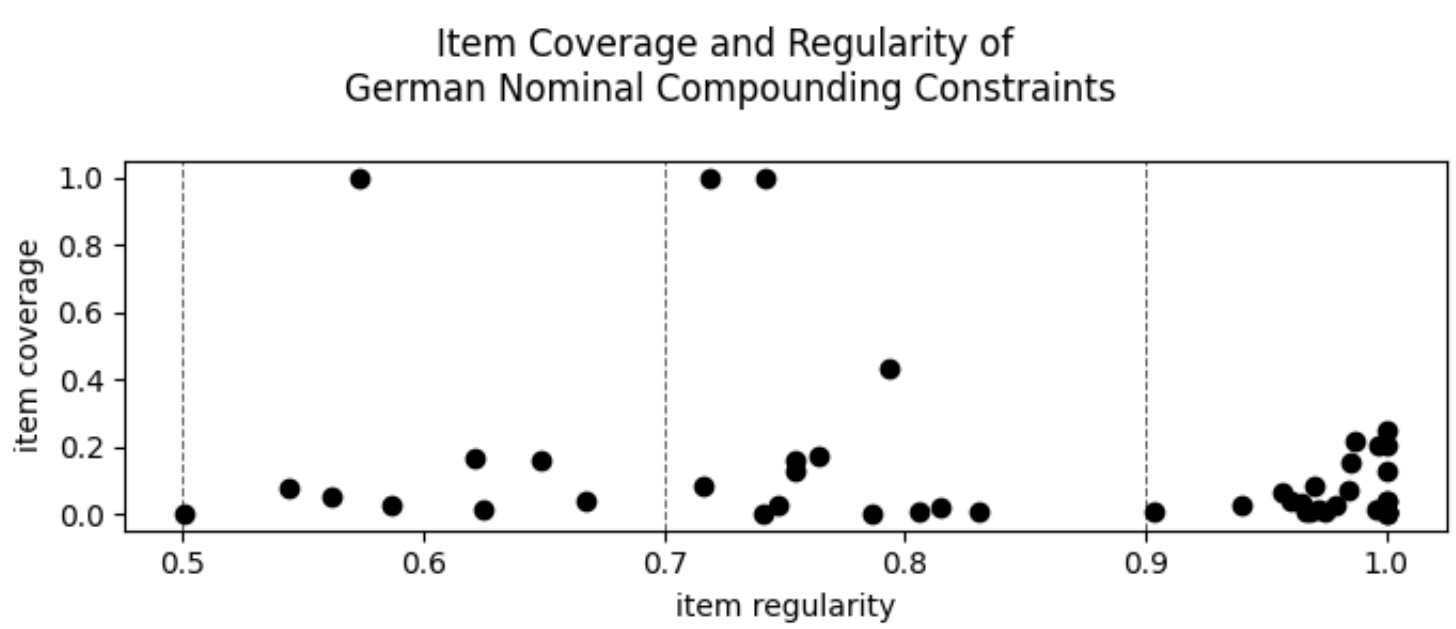
	N
total	49
p2l	32
l2p	16

1. As already reported in Shmalts 2026, the item coverage values are mostly very low, with more than half of the constraints lying in the item coverage range of $[\sim 0.0; 0.05)$$[\sim 0.0; 0.05)$ (28 of 49).

What concerns the item regularity values, the present study produces novel results. Specifically, it seems that the thresholds of $0.7$$0.7$ and $0.9$$0.9$ are the natural barriers between "low", "moderate", and "high" regularity. Indeed, the respective clusters are seen clearly on the scatter plot. The legitimacy of this values is further supported by the fact that they cut through the first and the second largest gaps between neighboring item regularity values in the dataset.

The item regularity values clustering calls the original regularity scale from Shmalts 2026 into question. We propose the newly discovered thresholds of $0.7$$0.7$ and $0.9$$0.9$ as the new (ir)regularity scale. More on completion of the new scale with a "regular" level follows in chapter 4.

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	N
cvg: [0.0-0.05)	28
cvg: [0.05-0.2)	13
cvg: [0.2-1.01)	8
reg: [0.5-0.7)	9
reg: [0.7-0.9)	13
reg: [0.9-1.01)	27

The two largest gaps betwee neighboring item regularity values are $0.831$$0.831$-$0.903$$0.903$ and $0.667$$0.667$-$0.716$$0.716$. They are cut precisely by the thresholds of $0.9$$0.9$ and $0.7$$0.7$, respectively.

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	left	right	diff
0	0.831	0.903	0.072
1	0.667	0.716	0.049
2	0.501	0.544	0.043
3	0.903	0.940	0.037
4	0.587	0.621	0.034
5	0.625	0.649	0.024
6	0.719	0.741	0.022
7	0.764	0.786	0.022
8	0.544	0.562	0.018
9	0.649	0.667	0.018

2. The L2P constraints appear to be statistically more regular than the P2L constraints. This conclusion corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirm its robustness after adding semantic constraints. On the contrary, there was no observed statistical difference between the item coverage values of the P2L vs. the L2P constraints, as unexpected as it was. This finding contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.

Almost all L2P applicability values have a very low coverage and a high regularity, while the applicability values of the P2L constraints are distributed more or less evenly. Thus, 12 out of 16, or $75\%$$75\%$ of the L2P constraints have an item coverage $<0.05$$<0.05$. For the P2L constraints, this number is only $50\%$$50\%$. The item regularity values are concentrated much closer to $1.0$$1.0$ among the L2P but not the P2L constraints. There are none L2P constraints with item regularity $<0.7$$<0.7$. On the contrary, there are eighr, or $25\%$$25\%$, such P2L constraints. Almost all L2P constraints are highly regular ($\geq 0.9$$\geq 0.9$): 13 out of 16, or $81.3\%$$81.3\%$. The P2L constraints do not even come close with mere $43.8\%$$43.8\%$ of highly regular constraints.

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	all	P2L	L2P
cvg: [0.0-0.05)	28	16	12
cvg: [0.05-0.2)	13	11	2
cvg: [0.2-1.01)	8	5	2
reg: [0.5-0.7)	9	8	0
reg: [0.7-0.9)	13	10	3
reg: [0.9-1.01)	27	14	13

We now run a Mann-Whitney U-test over the applicability values of the P2L vs. the L2P constraints. We choose the Mann-Whitney U-test because we are dealing with two independent samples whose distributions are abnormal and skewed. Since we suppose that the L2P constraints have lower coverage and higher regularity, we choose the two one-tailed alternative hypotheses: greater and less.

We can clearly see that the L2P constraints are undeniably more regular than the P2L constraints at sig = 0.001 (p-value $0.001$$0.001$; one-tailed alternative: L2P item regularity values are greater). They also have a lower coverage, with p-value of $0.017$$0.017$ (one-tailed alternative: P2L item coverage values are greater).

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		cvg	reg
alternative			
P2L>L2P	U	353.500	115.000
	p	0.017	0.999
P2L<L2P	U	353.500	115.000
	p	0.984	0.001

Since there are a few outliers in the P2L item coverage values, and *SP18_Pl_Constr* is a small dataset that can be not representative as of coverage, the results of the Mann-Whitney U-test can biased. We therefore additionally run a permutation test below. It compares the values themselves and not their ranks and is therefore much more sensitive to the outliers.

The permutation test confirmed that the item regularity values are higher among the L2P constraints (observed shift of the P2L vs. the L2P item regularity means $-0.131$$-0.131$, p-value $0.001$$0.001$). It also confirms our doubts about item coverage: the difference is not statistically different, with p-value of $\sim 0.06$$\sim 0.06$.

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		cvg	reg
alternative			
P2L>L2P	shift	0.088	-0.131
	p	0.060	0.999
P2L<L2P	shift	0.088	-0.131
	p	0.945	0.001

To demonstrate the bias of the outliers, we run a dedicated Mann-Whitney U-test over *GeCoDB_v07_GN* only with an item coverage threshold og $0.5$$0.5$, which is still much. While the robust regularity difference stays significant at p-value $0.003$$0.003$, the significance of the coverage difference vanishes (p-value $0.072$$0.072$). We conclude that there is no difference in item coverage between the P2L and the L2P constraints, after all.

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		cvg	reg
alternative			
P2L>L2P	U	235.500	89.000
	p	0.072	0.997
P2L<L2P	U	235.500	89.000
	p	0.932	0.003

3. Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho $-0.184$$-0.184$ and $-0.342$$-0.342$, respectively). Statistically, constraints with lower coverage have slightly higher regularity. This is in line with our assumption that the more specific constraints are stronger than the general ones. The large difference in the correlation coefficient between the item and type variants is a result of different productivity of items what regards building compounds.

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	item	type
r	-0.184	-0.342

4. We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding. Note that all non-confirmed constraints (item regularity $<0.5$$<0.5$) are semantic as well which makes the real picture even worse for semantics.

At this step, we will be treating the domain type of constraints as one-hot vectors. For example, a morphological+phonological constraint will be encoded as $[1, 0, 1, 0, 0]$$[1, 0, 1, 0, 0]$, which is decoded as `[+morphological][+derivational][+phonological][+semantic][+lexical]`. These vectors will then be passed to a linear regression model with an Ordinary Least Squares (OLS) loss function.

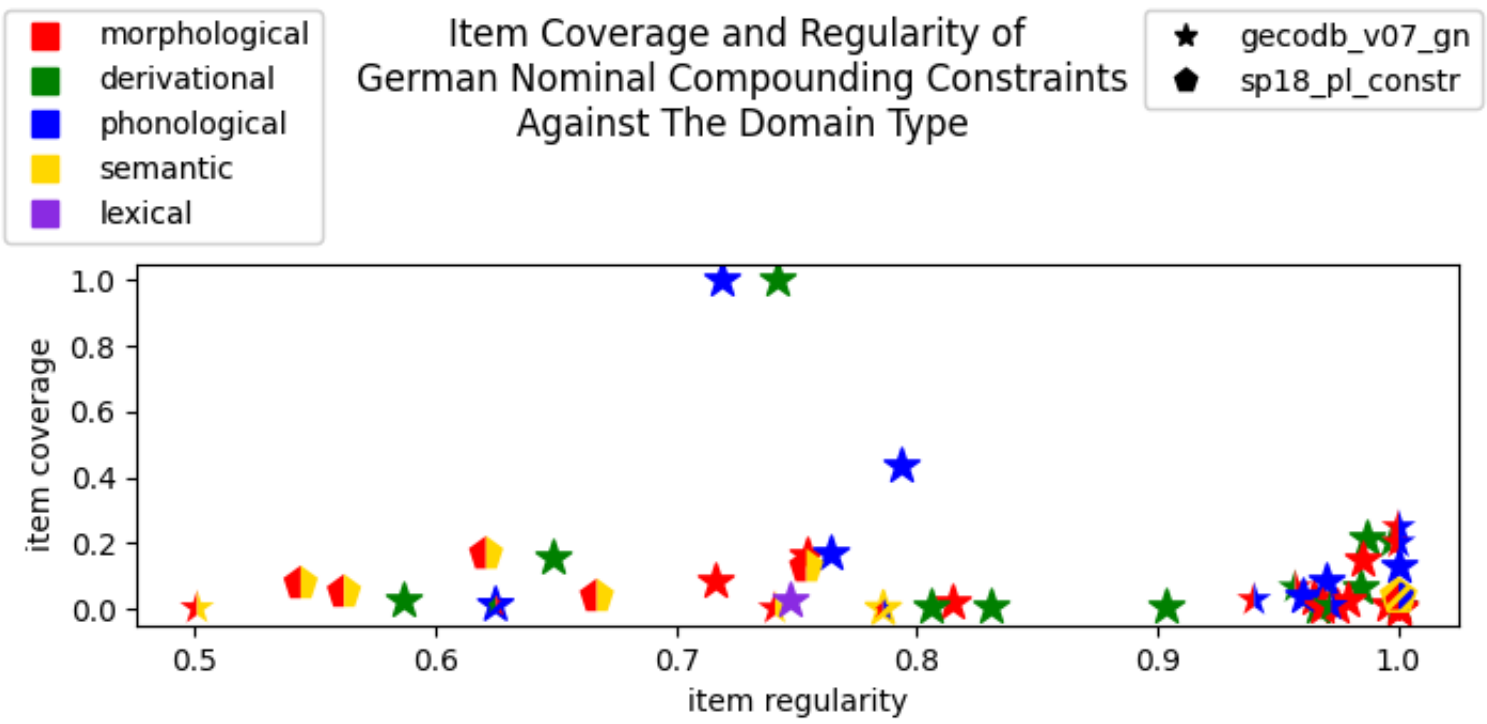
Here, we first create the vectors.

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	dataset	appl_item	n_appl_items	quantifier	cvg_item_abs	cvg_item	cvg_type	reg_item	reg_type	n_vio
	ecodb_v07_gn	n1	3319.0	always	29.0	0.009	0.008	0.966	0.833	
	ecodb_v07_gn	n1	3319.0	mostly	94.0	0.028	0.027	0.587	0.624	
	p18_pl_constr	n1+sem	78.0	irregularly	4.0	0.051	0.010	0.562	0.864	
	ecodb_v07_gn	n1+linker	3585.0	most	31.0	0.009	0.014	0.903	0.914	
	ecodb_v07_gn	n1	3319.0	tend	1443.0	0.435	0.508	0.793	0.709	

Let us consider the scatter plot of constraint item regularity against the domain type. We can immediately see that morphological, derivational, and phonological constraints are scattered all over the lot but are still concentrated towards the right edge of the axis. Contrary to that, semantic (including mixed) constraints only populate the range of $[0.5; 0.8]$$[0.5; 0.8]$; the only constraint with a high regularity is morphological+phonological+semantic. The plot suggests that the constraints that include a semantic condition are less regular.

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To confirm the intuition, we run a linear regression over the item regularity of the constraints vs. their domain type, based on the one-hot vectors we produced above. To concentrate on individual contribution of each domain, we do not include their interaction terms.

The linear regression model confirms our intuition. Semantics is the only domain that contributes to item regularity significantly (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$). That is, the item regularity of an arbitrary constraint decreases by $0.21$$0.21$ when a semantic condition included, assuming that the other domains of this constraint remain intact. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.289
Model:                            OLS   Adj. R-squared:                  0.205
Method:                 Least Squares   F-statistic:                     3.418
Date:                Tue, 11 Aug 2026   Prob (F-statistic):             0.0111
Time:                        16:51:33   Log-Likelihood:                 30.094
No. Observations:                  48   AIC:                            -48.19
Df Residuals:                      42   BIC:                            -36.96
Df Model:                           5
Covariance Type:            nonrobust
============================================================================================
                               coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------
Intercept                    0.8499      0.061     13.936      0.000       0.727       0.973
morphological[T.True]        0.0643      0.058      1.109      0.274      -0.053       0.181
derivational[T.True]  -6.965e-05      0.058     -0.001      0.999      -0.116       0.116
phonological[T.True]         0.0540      0.048      1.122      0.268      -0.043       0.151
semantic[T.True]            -0.2074      0.055     -3.742      0.001      -0.319      -0.096
lexical[T.True]             -0.1029      0.151     -0.681      0.499      -0.408       0.202
==============================================================================
Omnibus:                        2.435   Durbin-Watson:                   1.479
Prob(Omnibus):                  0.296   Jarque-Bera (JB):                1.949
Skew:                          -0.494   Prob(JB):                        0.377
Kurtosis:                       2.990   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

If we include the interaction terms, the picture stays the same. Now morphology, derivation, phonology, and semantics (but not the lexical domain) significantly contribute to item regularity (p-value in $[0.0; 0.001]$$[0.0; 0.001]$). From these four isolated domains, only semantics has a negative coefficient ($-0.128$$-0.128$) which the former three increase item regularity by $0.274$$0.274$, $0.174$$0.174$, $0.196$$0.196$, respectively (intercept $0.672$$0.672$). From all the interaction terms, the interaction of morphology and semantics is significant (coefficient $-0.128$$-0.128$, p-value $0.0$$0.0$). Otherwise, the interaction morphology+derivation+phonology is significant, too (coefficient $-0.352$$-0.352$, p-value $0.034$$0.034$) but we discard it since there is only one such constraint, which cannot assure statistical power.

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.469
Model:                            OLS   Adj. R-squared:                  0.344
Method:                 Least Squares   F-statistic:                     3.734
Date:                Tue, 11 Aug 2026   Prob (F-statistic):            0.00195
Time:                        16:51:33   Log-Likelihood:                 37.106
No. Observations:                  48   AIC:                            -54.21
Df Residuals:                      38   BIC:                            -35.50
Df Model:                           9
Covariance Type:            nonrobust
================================================================================================
                                                   c
------------------------------------------------------------------------------------------------
Intercept                                          0.6
morphological[T.True]                              0.2
derivational[T.True]                               0.1
phonological[T.True]                               0.1
semantic[T.True]                                  -0.1
lexical[T.True]                                    0.0
morphological[T.True]:derivational[T.True]        -0.1
morphological[T.True]:phonological[T.True]        -0.1
derivational[T.True]:phonological[T.True]         -0.0
morphological[T.True]:semantic[T.True]            -0.1
derivational[T.True]:semantic[T.True]
phonological[T.True]:semantic[T.True]              0.0
morphological[T.True]:lexical[T.True]
derivational[T.True]:lexical[T.True]
phonological[T.True]:lexical[T.True]
semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]   -0.3
morphological[T.True]:derivational[T.True]:semantic[T.True]
morphological[T.True]:phonological[T.True]:semantic[T.True]        0.0
derivational[T.True]:phonological[T.True]:semantic[T.True]
morphological[T.True]:derivational[T.True]:lexical[T.True]
morphological[T.True]:phonological[T.True]:lexical[T.True]
derivational[T.True]:phonological[T.True]:lexical[T.True]
morphological[T.True]:semantic[T.True]:lexical[T.True]
derivational[T.True]:semantic[T.True]:lexical[T.True]
phonological[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:semantic[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
derivational[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
==============================================================================
Omnibus:                        0.912   Durbin-Watson:                   1.901
Prob(Omnibus):                  0.634   Jarque-Bera (JB):                0.275
Skew:                          -0.055   Prob(JB):                        0.872
Kurtosis:                       3.354   Cond. No.                     1.46e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 3.97e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

The domain tyoe of constraints has no effect on their item coverage.

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                cvg_item   R-squared:                       0.117
Model:                            OLS   Adj. R-squared:                  0.012
Method:                 Least Squares   F-statistic:                     1.110
Date:                Tue, 11 Aug 2026   Prob (F-statistic):              0.370
Time:                        16:51:33   Log-Likelihood:                 10.975
No. Observations:                  48   AIC:                            -9.951
Df Residuals:                      42   BIC:                             1.277
Df Model:                           5
Covariance Type:            nonrobust
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                        0.2433      0.091      2.678      0.011       0.060       0.427
morphological[T.True]           -0.1578      0.086     -1.829      0.075      -0.332       0.016
derivational[T.True]            -0.1036      0.086     -1.208      0.234      -0.277       0.069
phonological[T.True]             0.0132      0.072      0.184      0.855      -0.131       0.158
semantic[T.True]                -0.0338      0.083     -0.409      0.684      -0.200       0.133
lexical[T.True]                 -0.2173      0.225     -0.966      0.340      -0.671       0.237
==============================================================================
Omnibus:                       54.471   Durbin-Watson:                   0.802
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              276.619
Skew:                           3.002   Prob(JB):                     8.57e-61
Kurtosis:                      13.113   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```